

LAPORAN
WORKSHOP SISTEM OPERASI



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Percobaan1

```
suci@UbuntuSuciRamadhani2SIC:~$ su -
Password:
root@UbuntuSuciRamadhani2SIC:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defau
lt qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:6e:f3:0d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 85185sec preferred_lft 85185sec
    inet6 fd17:625c:f037:2:8388:c921:b47a:a3b4/64 scope global temporary dynami
c
        valid_lft 85915sec preferred_lft 13915sec
    inet6 fd17:625c:f037:2:2d67:8c85:7bd5:25d2/64 scope global dynamic mngtmpad
r noprefixroute
        valid_lft 85915sec preferred_lft 13915sec
    inet6 fe80::fcf4:6d97:3255:7226/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
root@UbuntuSuciRamadhani2SIC:~# ip route
default via 10.0.2.2 dev enp0s3 proto dhcp metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 scope link metric 1000
r Show Applications amadhani2SIC:~#
```

Pertama saya memasuki root

Ip a: *Di sini saya mengecek interface mana saja yang aktif, IP apa saja yang sedang digunakan, serta memastikan NIC utama (misalnya **enp0s3**) sudah terdeteksi.*

ip route: *Saya melihat rute default — ini penting untuk mengetahui IP gateway yang digunakan sistem untuk akses internet.*

```
Global
LLMNR setting: no
MulticastDNS setting: no
DNSOverTLS setting: no
DNSSEC setting: no
DNSSEC supported: no
DNSSEC NTA: 10.in-addr.arpa
             16.172.in-addr.arpa
             168.192.in-addr.arpa
             17.172.in-addr.arpa
             18.172.in-addr.arpa
             19.172.in-addr.arpa
             20.172.in-addr.arpa
             21.172.in-addr.arpa
             22.172.in-addr.arpa
             23.172.in-addr.arpa
             24.172.in-addr.arpa
             25.172.in-addr.arpa
             26.172.in-addr.arpa
             27.172.in-addr.arpa
             28.172.in-addr.arpa
             29.172.in-addr.arpa
             30.172.in-addr.arpa
             31.172.in-addr.arpa
             corp
             d.f.ip6.arpa
             home
             internal
```

lines 1-28

systemd-resolve --status: *Saya memastikan server DNS sudah ada, karena tanpa DNS ping domain ga akan berhasil.*

Percobaan2

```
GNU nano 4.8 /etc/netplan/01-network-manager-all.yaml Modify
# Let NetworkManager manage all devices on this system
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    enp0s3:
      dhcp4: true

^G Get Help      ^O Write Out    ^W Where Is     ^K Cut Text     ^J Justify
^X Exit          ^R Read File    ^_ Replace      ^U Paste Text   ^T To Spell
```

`sudo nano /etc/netplan/01-network-manager-all.yaml`: Saya membuka file konfigurasi jaringan untuk mengatur mode DHCP.

```
root@UbuntuSuciRamadhani2SIC:~# sudo netplan apply
root@UbuntuSuciRamadhani2SIC:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:6e:f3:0d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86396sec preferred_lft 86396sec
    inet 10.0.2.50/24 brd 10.0.2.255 scope global secondary noprefixroute enp0s3
    inet6 fe80::a00:27ff:fe6e:f30d/64 scope link
        valid_lft forever preferred_lft forever
```

sudo netplan apply

ip a

Setelah apply, saya cek apakah IP sudah diberikan otomatis oleh DHCP server.

Percobaan3

```
GNU nano 4.8 /etc/netplan/01-network-manager-all.yaml Modified
# Let NetworkManager manage all devices on this system
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    enp0s3:
      dhcp4: true
      addresses:
        - 10.0.2.255
      gateway4: 10.0.2.255
      nameservers:
        addresses:
          - 8.8.8.8
          - 1.1.1.1
```

^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify
^X Exit ^R Read File ^\ Replace ^U Paste Text ^T To Spell

`sudo nano /etc/netplan/01-network-manager-all.yaml`: Kali ini saya mengubah konfigurasi agar interface memakai IP static.

```
root@UbuntuSuciRamadhani2SIC:~# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=9.93 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=11.6 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=10.2 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=255 time=10.1 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=255 time=12.7 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=255 time=10.0 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=255 time=10.9 ms
^Z
[2]+  Stopped                  ping 8.8.8.8
```

`sudo netplan apply`

`ip a`

`ping 8.8.8.8`

Saya memastikan IP static sudah muncul, dan mengetes apakah IP static tersebut bisa akses internet.

```
root@UbuntuSuciRamadhani2SIC:~# ping google.com
PING google.com (64.233.170.102) 56(84) bytes of data.
64 bytes from sg-in-f102.1e100.net (64.233.170.102): icmp_seq=1 ttl=255 time=12
.0 ms
64 bytes from sg-in-f102.1e100.net (64.233.170.102): icmp_seq=2 ttl=255 time=10
.7 ms
64 bytes from sg-in-f102.1e100.net (64.233.170.102): icmp_seq=3 ttl=255 time=11
.4 ms
64 bytes from sg-in-f102.1e100.net (64.233.170.102): icmp_seq=4 ttl=255 time=10
.6 ms
^Z
[3]+  Stopped                  ping google.com
```

ping google.com: *Jika ini berhasil, berarti DNS server bekerja dengan baik.*

```
root@UbuntuSuciRamadhani2SIC:~# ip a show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:6e:f3:0d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86289sec preferred_lft 86289sec
    inet 10.0.2.50/24 brd 10.0.2.255 scope global secondary noprefixroute enp0s
3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe6e:f30d/64 scope link
        valid_lft forever preferred_lft forever
```

ip a show enp0s3: *Saya memastikan interface dasar yang akan jadi “trunk” VLAN.*

```
root@UbuntuSuciRamadhani2SIC:~# sudo apt install vlan
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  vlan
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 11.3 kB of archives.
After this operation, 50.2 kB of additional disk space will be used.
Get:1 http://id.archive.ubuntu.com/ubuntu focal-updates/universe amd64 vlan all
2.0.4ubuntu1.20.04.1 [11.3 kB]
Fetched 11.3 kB in 0s (108 kB/s)
```

```

Processing triggers for man-db (2.9.1-1) ...
root@UbuntuSuciRamadhani2SIC:~# sudo modprobe 8021q
root@UbuntuSuciRamadhani2SIC:~# sudo ip link add link enp0s3 name enp0s3.10 type
vlan id 10
root@UbuntuSuciRamadhani2SIC:~# sudo ip addr add 10.0.10.50/24 dev enp0s3.10
root@UbuntuSuciRamadhani2SIC:~# sudo ip link set enp0s3.10 up
root@UbuntuSuciRamadhani2SIC:~# sudo ip link add link enp0s3 name enp0s3.20 typ
e vlan id 20
root@UbuntuSuciRamadhani2SIC:~# sudo ip addr add 10.0.20.50/24 dev enp0s3.20
root@UbuntuSuciRamadhani2SIC:~# sudo ip link set enp0s3.20 up
root@UbuntuSuciRamadhani2SIC:~#

```

sudo apt install vlan

sudo modprobe 8021q

Ini mengaktifkan kemampuan Linux membuat interface VLAN.

Membuat VLAN 10

sudo ip link add link enp0s3 name enp0s3.10 type vlan id 10

sudo ip addr add 10.0.10.50/24 dev enp0s3.10

sudo ip link set enp0s3.10 up

*Saya membuat interface bernama **enp0s3.10** (dot1q tag 10), memberikan IP untuk VLAN 10, lalu mengaktifkannya.*

Membuat VLAN 20

sudo ip link add link enp0s3 name enp0s3.20 type vlan id 20

sudo ip addr add 10.0.20.50/24 dev enp0s3.20

sudo ip link set enp0s3.20 up

Sama seperti VLAN 10, tetapi ini khusus VLAN 20.

```

root@UbuntuSuciRamadhani2SIC:~# ip a show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:6e:f3:0d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 85625sec preferred_lft 85625sec
    inet 10.0.2.50/24 brd 10.0.2.255 scope global secondary noprefixroute enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe6e:f30d/64 scope link
        valid_lft forever preferred_lft forever
root@UbuntuSuciRamadhani2SIC:~# ping 10.0.10.50
PING 10.0.10.50 (10.0.10.50) 56(84) bytes of data.
64 bytes from 10.0.10.50: icmp_seq=1 ttl=64 time=0.023 ms
64 bytes from 10.0.10.50: icmp_seq=2 ttl=64 time=0.079 ms
64 bytes from 10.0.10.50: icmp_seq=3 ttl=64 time=0.039 ms
^Z
[4]+  Stopped                  ping 10.0.10.50
root@UbuntuSuciRamadhani2SIC:~# ping 10.0.20.50
PING 10.0.20.50 (10.0.20.50) 56(84) bytes of data.
64 bytes from 10.0.20.50: icmp_seq=1 ttl=64 time=0.020 ms
64 bytes from 10.0.20.50: icmp_seq=2 ttl=64 time=0.035 ms
^Z
[5]+  Stopped                  ping 10.0.20.50

```

ip a show enp0s3: Saya melihat apakah interface **enp0s3.10** dan **enp0s3.20** sudah tampil.

ping 10.0.10.50

ping 10.0.20.50

Jika ini berhasil, berarti kedua VLAN aktif.

```

root@UbuntuSuciRamadhani2SIC:~# sudo sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
root@UbuntuSuciRamadhani2SIC:~# ping 8.8.8.8 -I enp0s3.10
PING 8.8.8.8 (8.8.8.8) from 10.0.10.50 enp0s3.10: 56(84) bytes of data.
From 10.0.10.50 icmp_seq=1 Destination Host Unreachable
From 10.0.10.50 icmp_seq=2 Destination Host Unreachable
From 10.0.10.50 icmp_seq=3 Destination Host Unreachable
^Z
[7]+  Stopped                  ping 8.8.8.8 -I enp0s3.10
root@UbuntuSuciRamadhani2SIC:~# ping 8.8.8.8 -I enp0s3.20
PING 8.8.8.8 (8.8.8.8) from 10.0.20.50 enp0s3.20: 56(84) bytes of data.
From 10.0.20.50 icmp_seq=1 Destination Host Unreachable
From 10.0.20.50 icmp_seq=2 Destination Host Unreachable
From 10.0.20.50 icmp_seq=3 Destination Host Unreachable
^Z
[ Show Applications          ping 8.8.8.8 -I enp0s3.20

```

sudo sysctl -w net.ipv4.ip_forward=1

Saya menghidupkan fungsi router pada Linux supaya paket antar VLAN bisa diproses dan diarahkan ke jalur yang benar.

ping 10.0.20.50 -I enp0s3.10

*Saya mencoba memaksa ping keluar dari VLAN 10 menuju VLAN 20.
Tanpa routing, hasilnya biasanya gagal — dan ini normal.*

ping 8.8.8.8 -I enp0s3.10

ping 8.8.8.8 -I enp0s3.20

Saya memastikan bahwa masing-masing VLAN bisa keluar ke internet menggunakan interface mereka masing-masing.

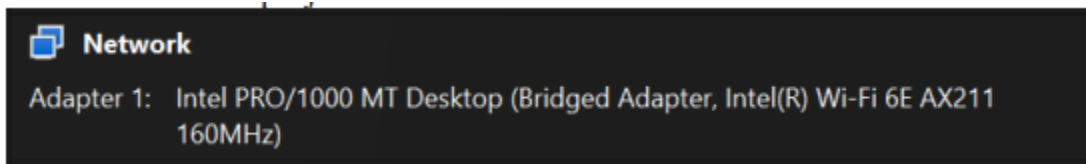
LATIHAN

1. Lakukan analisis mengapa ketika melakukan ping antara VLAN10 dan VLAN20 terjadi keterangan:

```
From 192.168.10.101 icmp_seq=1 Destination Host Unreachable
From 192.168.10.101 icmp_seq=2 Destination Host Unreachable
From 192.168.10.101 icmp_seq=3 Destination Host Unreachable
From 192.168.10.101 icmp_seq=4 Destination Host Unreachable
```

- Ping antar VLAN gagal karena VLAN10 dan VLAN20 berada di jaringan yang berbeda dan terisolasi, sehingga tidak ada jalur (routing) yang menghubungkan keduanya.
- Secara default, Linux tidak melakukan routing antar subnet. Akibatnya, saat VLAN10 mencoba ping VLAN20, paket tidak tahu harus lewat mana.

2. Silahkan cari solusi agar bisa melakukan ping antar VLAN, nanti hasilnya silahkan di screenshot dan beri penjelasan.



Network
Adapter 1: Intel PRO/1000 MT Desktop (Bridged Adapter, Intel(R) Wi-Fi 6E AX211 160MHz)

- Pakai Host-only = tidak bisa
- Pakai NAT tidak bisa
- Pakai bridged adapt = tidak bisa

Berdasarkan searching, hasilnya Wi-Fi TIDAK mendukung VLAN (802.1Q) VirtualBox tidak bisa melewatkan frame VLAN melalui WiFi, sehingga:

- VLAN 10 dan VLAN 20 memang bisa muncul di Ubuntu
- TAPI paket VLAN tidak pernah keluar NIC
- Semua ping antar VLAN akan tetap → Destination Host Unreachable

Ini bukan salah konfigurasi, tapi keterbatasan hardware WiFi. Solusi yang disarankan :

- Gunakan Ethernet LAN : Colok kabel LAN ke laptop → VirtualBox → Network → Bridged Adapter → pilih Ethernet.
- Gunakan USB-to-LAN Adapter
- Kerjakan di komputer LAB : Komputer lab pakai LAN → VLAN pasti berjalan.
- Haru